

Certified Lattice Data and a Within-Family Selector for the Cooper s_7/s_{10} $K3$ Families

Xavier Callens
SocrateAI Scientific Agora Collaboration

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What is not claimed

This is a mathematics note about lattice arithmetic on two families of $K3$ surfaces. It makes no physical claim of any kind: no dark-matter identification, no dark-energy statement, no observable. The programme’s flux/tadpole route is blocked pending a specified threefold base (F5b) and a selection among the surfaces discussed here maps to no measurement. These renunciations are load-bearing, not boilerplate; see section 1.

Abstract

We report a session’s results on the Cooper s_7 and s_{10} Picard–Fuchs families, all of it lattice mathematics with an explicit epistemic tier on every claim. Six points of the $\rho = 20$ locus of each family’s transcendental lattice are exhibited as exact vectors with saturated orthogonal complements, reduced binary forms, and frame indices; the lattice arithmetic of all twelve is kernel-checked in Lean 4, three of the six *independently* by two separate developments, which we re-verify ourselves before citing either. A within-family selector — minimal discriminant of the transcendental lattice — is adopted by an explicit ruling among two computed alternatives, chosen because the rejected alternative is not unique in one family. An explicit M_7 -polarized model realizes the s_7 family as an Inose surface; its elliptic fibration, taken from a sourced literature equation and re-derived here, has the two singular loci of the Picard–Fuchs operator indistinguishable at the level of discriminant-root orders, which answers a standing question about one of them in the negative. Stage-2 monodromy of the order-two operator underlying each family is certified by rigorous ball arithmetic, replacing an earlier numerical-tolerance heuristic with enclosures whose diameters are reported exactly. We record, in band, five errors made and caught during the same session, because the standing discipline of this programme requires retractions to be machine-visible rather than a footnote.

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1 Scope and what is not claimed

What is not claimed

No claim in this note is physical. The $\rho = 20$ locus is a set of lattice points on a modular curve; it maps to no observable and licenses no dark-sector identification. The discriminant-floor selector of section 3 is a naming convention for which candidate a certificate calls “distinguished” within a family, adopted by an explicit ruling, and it does not rank one Cooper family against the other. No Kodaira fibre type is attached to any point of either family’s Picard–Fuchs operator: that reading is a category error for this class of family (elliptic points of a modular curve are not Kodaira degenerations), and the one place an explicit Weierstrass model is computed (section 4.1) we report discriminant-root *orders* only, withholding the label pending a separate ruling. The s_{10} family’s transcendental lattice remains a DRAFT certificate; every s_{10} result here is advisory and does not depend on, or promote, that certificate’s status.

2 The six $\rho = 20$ locus rows, kernel-checked

Each register family (s_7 at level $N = 7$, s_{10} at level $N = 10$) carries a rank-three lattice $U \oplus \langle 2N \rangle$ with quadratic form $v^2 = 2xy + 2Nz^2$ on coordinates $v = (x, y, z)$. Three points per family are singled out by the vanishing locus of the family’s order-three Picard–Fuchs operator L_3 : the two finite singular points and the point at infinity. At each such point the transcendental lattice, rank 19 generically, drops to rank 2 and becomes positive definite — ρ jumps to 20. Table 1 lists all six, as exact vectors with their norm, the exact orthogonal complement (its Gram matrix, reduced to a binary quadratic form of discriminant D), and the Lean kernel status.

Family	z	v	v^2	D	T (reduced)	Tier / sources
cooper_s7	−1	(2, −4, 1)	−2	−7	[2, 1; 1, 4]	Tier A, 2 source(s)
cooper_s7	1/27	(1, −1, 0)	−2	−28	[2, 0; 0, 14]	Tier A, 2 source(s)
cooper_s7	∞	(14, −14, −5)	−42	−3	[2, 1; 1, 2]	Tier A, 2 source(s)
cooper_s10	−1/4	(2, −6, 1)	−4	−20	[4, 2; 2, 6]	Tier A, 1 source(s)
cooper_s10	1/16	(1, −1, 0)	−2	−40	[2, 0; 0, 20]	Tier A, 1 source(s)
cooper_s10	∞	(10, −10, −3)	−20	−4	[2, 0; 0, 2]	Tier A, 2 source(s)

Table 1: The six locus rows. **Tier A — Lean kernel** for the lattice arithmetic (norm, saturated complement, Gram, reduced form, frame determinant) of every row; three rows carry two independent kernel developments. **Tier B — derived, not measured**: the numeric recognition of z , and the identification of the orthogonal complement with the family’s transcendental lattice T_X (Dolgachev 1996 §7, Doran 1998 Thm. 5.13, **Tier L — literature**). The index step $|\det(\text{frame})| = |v^2|/\text{div}(v)$ is checked per row, never proved in general, in either kernel development.

Headline result

For the two rows at $z = 1/27$ and $z = -1$ (the s_7 family) and, independently added this session, both families’ $z = \infty$ rows, the lattice half is kernel-checked *twice*, by two developments written without reading each other’s source, then read at the statement level and re-gated by us in each producer’s own build tree before either is cited.

Remark 2.1. The two independent developments state the norm, the exact orthogonal complement as a set (an if-and-only-if, not merely two orthogonal vectors), the complement’s Gram matrix, its Gauss-reduced form, and the frame determinant, each as a standalone kernel-checked declaration, over the standard axiom set `{propext, Classical.choice, Quot.sound}` and nothing else. Producer and verifier are different in every case reported here: we re-ran the build, the sorry grep, the axiom audit, the statement lock, and `#print axioms` ourselves, in the producer’s own worktree, before writing any lattice tier into a certificate.

3 An adopted selector, and the alternative it rejected

Six candidates are too many to call “the” surface of a family without saying which quantity is being extremized. Two computed candidates were compared:

- **SEL-D**: minimal $|\text{disc } T|$ within a family, verified unique not merely by class number (which counts reduced forms) but by the count of points *on the modular curve* itself.
- **SEL-N**: minimal $|v^2|$ within a family — the -2 floor achieved by an N -uniform Lean theorem in both kernel developments.

Table 2 gives both, exactly, per family.

Family	SEL-D (min $ \text{disc } T $)	SEL-N (min $ v^2 $)	Agree?
<code>cooper_s7</code>	$T = [2, 1; 1, 2], D = -3$	tie: $D = -7, D = -28$ (2 rows)	No
<code>cooper_s10</code>	$T = [2, 0; 0, 2], D = -4$	$D = -40$ (unique)	No

Table 2: SEL-D and SEL-N disagree in both families. SEL-N is *not unique* for s_7 : two rows tie at the -2 floor, corresponding to the two fixed points of the Atkin–Lehner involution W_7 on the family’s Hauptmodul coordinate.

SEL-D was adopted. The choice was put, with both alternatives computed and the disagreement stated, to the programme’s designated decision authority, who selected SEL-D directly. This narrows an earlier, separately-ruled clause (“no minimum-discriminant rule”) to cross-family comparisons only; it does not rank s_7 against s_{10} , and it does not promote s_{10} ’s DRAFT lattice certificate. The selected surfaces — $T = A_2$ ($D = -3$) for s_7 , $T = \langle 2 \rangle \oplus \langle 2 \rangle$ ($D = -4$) for s_{10} , advisory — extremize the same quantity that a separate, pre-existing attractor-mechanism development in this programme’s Lean corpus already names for the smallest admissible black-hole charge configuration; both are instances of the same discriminant-gap floor ($4ac - b^2 \geq 3$), so the coincidence is forced by that shared floor and is stated once, here, as an observation and not as corroborating evidence for either result.

4 An explicit M_7 -polarized model

Restricting the Clingher–Doran–Lewis–Whitcher normal form for M -polarized $K3$ surfaces to the M_7 -polarized locus realizes the s_7 family as an Inose surface $X(a(z), b(z))$, with $a^3 = \pi(z)$,

$$b^2 = \pi(z) - \sigma(z) + 1 \text{ and}$$

$$\sigma(z) = J(\tau) + J(7\tau), \quad \pi(z) = J(\tau) J(7\tau),$$

where z is the family’s own $\Gamma_0(7)^+$ Hauptmodul coordinate and J is Klein’s J -invariant, normalized $J(i) = 1$. Because $X_0(7)^+$ has a single cusp, σ and π are Laurent polynomials in z ; both were solved from their q -expansions and verified against every held-out coefficient to order q^{128} (Tier A — Lean kernel for the finite-order verification; Tier B — derived, not measured for the identification of z with the Hauptmodul coordinate). Exactly,

$$\sigma(z)^2 - 4\pi(z) = \frac{-(z+1)(27z-1)(2z-1)^2(8z-1)^2(24z-1)^2(8z^2+16z-1)^2}{2985984 z^{14}}.$$

The *simple* zeros of this expression are exactly the two finite singular loci of L_3 ($z = -1$, $D = -7$; $z = 1/27$, $D = -28$): the points where the two isogenous elliptic curves become isomorphic, which is where $J(\tau) - J(7\tau)$ behaves like a square root, forced by the involution that swaps the two curves. The double zeros are the other self-7-isogenous CM points in the same window, each occurring exactly $h(D)$ times.

4.1 The fibration

A degree-two base change of a sourced elliptic pencil on $\text{Km}(E_1 \times E_2)$ (Kuwata–Shioda, §5.3, Tier L — literature) gives the Inose fibration explicitly, with the transcription checked against the source’s own printed discriminant before any further computation. Table 3 gives the discriminant-root orders — never a Kodaira label, per section 1 — for the generic case, both s_7 singular loci, and the two special values $J = 0, 1$.

Case	Discriminant-root orders	Note
Generic ($E_1 \not\cong E_2$)	$\{10, 10, 1, 1, 1, 1\}$	—
cooper_s7, $z = -1$	$\{10, 10, 2, 1, 1\}$	$J = -125/64$
cooper_s7, $z = 1/27$	$\{10, 10, 2, 1, 1\}$	$J = 614125/64$
$J = 1$ ($E_1 \cong E_2$)	$\{10, 10, 2, 2\}$	—
$J = 0$ ($E_i \cong E_\omega$)	$\{10, 10, 4\}$	—

Table 3: Discriminant-root orders of the Inose fibration. The mechanism behind the middle two rows: when the two elliptic curves become isomorphic, the discriminant’s constant term vanishes identically, and after the base change the two simple roots collapse onto the branch point, producing exactly one order-2 root there.

Headline result

The two singular loci of s_7 are *indistinguishable* at the level of discriminant-root orders: both give $\{10, 10, 2, 1, 1\}$. A standing hypothesis in this programme — that the two loci differ in their fibre configuration — is answered in the negative by this computation, for this fibration. The second algebraic class responsible for the rank jump at either point must therefore be a Mordell–Weil section, not a reducible fibre.

5 Certified stage-two monodromy

The order-two Picard–Fuchs operator underlying each family’s Sym^2 structure has monodromy that was previously recognized only by a numeric-tolerance heuristic (60-digit precision, a 10^{-35} gate). This session replaced that heuristic with rigorous ball arithmetic (Arb, via `python-flint`):

Frobenius series tails and Taylor-continuation tails are bounded by majorant induction on the operator’s own recurrence, giving enclosures whose diameters certify that each recognized monodromy entry is the *unique* rational of bounded denominator inside its enclosure. Table 4 reports the diameters achieved; the certified matrices were then fed into the exact lattice pipeline and reproduce every field of the corresponding lattice certificate, closing the chain from certified numerics to exact lattice arithmetic.

Family	Clause	Enclosure diameter
<code>cooper_s7</code>	$z = -1$	1.6×10^{-96}
<code>cooper_s7</code>	$z = 1/27$	1.0×10^{-115}
<code>cooper_s10</code>	$z = -1/4$	4.2×10^{-102}
<code>cooper_s10</code>	$z = 1/16$	2.2×10^{-115}

Table 4: Enclosure diameters of the certified stage-two monodromy. All are many orders of magnitude below the denominator bound required for the recognized rational to be unique.

6 Corrections recorded in band

Corrections recorded in this session

Five errors were made and caught within this session, listed because the standing discipline of this programme requires a retraction to be machine-visible, not a prose footnote.

1. An initial claim that SEL-N (section 3) was unique and level-uniform was false: two rows of the s_7 family tie at the -2 norm floor. Caught before any recommendation was written down.
2. An initial reading of a standing “no minimum-discriminant rule” as scoped to cross-family comparison only was a stretch of the text, not what it said; the question was put to the decision authority rather than resolved by inference.
3. A transcription of the Inose $(0,0)/(1,0)$ labels for two special points was swapped; corrected after an independent Weierstrass-reduction computation disagreed with it.
4. Two software defects were found while extending the Lean-attestation machinery of section 2: a worktree-path resolver that only recorded one build tree per repository, so a second, later commit on a different branch of the same repository silently failed its own file-hash check; and a test control that matched a lattice row by vector coordinates alone, which collide across the two register families.
5. A version-control error landed a pull request containing only a brief document, omitting the actual code and data changes its own description claimed; caught by comparing a certificate’s hash on the merged branch against the local file, and corrected by a follow-up commit before any downstream party was told the (wrong) hash.
6. *Dated correction, 2026-10-07.* The sentences of section 1 and section 3 calling the s_{10} lattice certificate a DRAFT, and the “(ADVISORY)” tags on the s_{10} rows of tables 1 and 2, were true when written on 2026-09-29 and were superseded later that day by a ruling that made the value-identical certificate LIVE; the certificates were re-emitted without the flag on 2026-10-07 and the two tables, which are generated from them, now show none. No value in any table changed. The prose sentences are left as written and are dated by this item; nothing in this paper ranks the two families, and that was never conditional on the flag.
7. *Dated correction, 2026-10-08.* Item 6 was itself partly false as built on 2026-10-07: the table generator attached “(ADVISORY)” to the s_{10} rows of table 1 by family name, not from a certificate field, so that build still printed the tag on three rows. The generator now reads the certificate’s `advisory_family` flag (false for every row) and has a control that fails if the tag is keyed on anything else; this rebuild carries no tag. The same fix replaced typed discriminant-root orders in the generator of table 3 by values read from the certificate (the printed values were right). The heading of this box says “five errors”; items 6 and 7 were added later, as dated corrections.

7 Reproducibility

Every number in tables 1 to 4 is generated directly from certificates in `data/certificates/` by `scripts/render_paper_tables.py`, which fails closed (`-check`) if any table fragment drifts from the certificates that produced it; nothing in this document’s tables is a typed literal. The checkers producing the underlying certificates each ship negative controls that must fail on a tampered input before the corresponding positive result is trusted. A full account of every

certificate, checker, and the exact gate output for both independent Lean developments of section 2 is in the repository’s decision and status records, referenced in section 8.

Provenance

Generated-by: Claude (Sonnet 5), Stream 2 | Verified-by: `scripts/render_paper_tables.py` -check; every cited checker’s own controls | Reviewed-by: N (preprint; not reviewed).

8 References and source records

References

- [1] I. Dolgachev, “Mirror symmetry for lattice polarized $K3$ surfaces,” *arXiv:alg-geom/9502005* (1996).
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- [4] A. Clingher, C. F. Doran, J. Lewis, and U. Whitcher, “Normal forms, $K3$ surface moduli, and modular parametrizations,” *arXiv:0712.1880*.
- [5] C. F. Doran, A. Harder, A. Y. Novoseltsev, and A. Thompson, “Families of lattice polarized $K3$ surfaces with monodromy,” *arXiv:1312.6434*.
- [6] M. Kuwata and T. Shioda, “Elliptic parameters and defining equations for elliptic fibrations on a Kummer surface,” *arXiv:math/0609473*.
- [7] G. Moore, “Arithmetic and Attractors,” *arXiv:hep-th/9807087* (1998).
- [8] SocrateAI Scientific Agora Collaboration, **SocrateAI-Scientific-Agora-K3-DarkMatter**, `K3_CRITERIA.md` and `TODO.md` at commit `d3a2a01` (2026-09-29); decision records `briefs/T0_DECISIONS_2026_09_28_STREAM2.md` (selector ruling), `briefs/STREAM2_AM8_SELECTOR_COMPARISON_2026_09_28.md` (comparison detail), `briefs/WP_GE10_INOSE_MODEL_M7_2026_09_27.md` (fibration derivation).